

Cross-Dataset Generalization in Pedestrian Crossing Intention

Prediction: A Diagnosis, a Data Artifact, and a Partial Fix

Abstract—Pedestrian crossing-intention predictors are almost universally evaluated by training and testing on the same dataset. Prior work showed this hides severe generalization failure: performance near chance when a model trained on one benchmark (PIE or JAAD) is tested, unchanged, on the other. No architecture paper we identify in the 2024–2026 literature re-evaluates this protocol. We revisit cross-dataset transfer for SF-GRU, a representative fusion architecture, and in the process discover that JAAD’s standard data loader silently mislabels the majority of samples under its default configuration – an artifact independent of, but entangled with, cross-dataset evaluation – which we identify, correct, and confirm does not itself explain the collapse. Under corrected labels, we run a falsification campaign against the two most natural explanations for the collapse and for a related split-sensitivity phenomenon: camera-geometry mismatch (tested via real metric-space ground-plane reprojection and a 20-point camera-parameter sensitivity sweep) and video-level train/test leakage (tested via a purpose-built video-grouped data split). Both are ruled out. Motivated by a modality-zeroing ablation that instead localizes sensitivity to the visual scene-context branch, we apply domain-adversarial training combined with a contrastive scale-invariance term, obtaining a real, seed-confirmed improvement in both transfer directions ($n = 6$ seeds; PIE→JAAD AUC 0.498 → 0.621, JAAD→PIE AUC 0.402 → 0.613) that nonetheless leaves cross-dataset performance well below in-domain levels. Using an independent representation probe, we then show this improvement is *dissociated* from the literal domain-invariance the method is designed to induce: the configurations that improve transfer AUC also make the fused representation *more*, not less, separable by an external probe, while a control that reduces separability also reduces AUC. Testing the same recipe on two further, structurally distinct fusion architectures shows it is not a general-purpose fix: on box-anchored cross-attention, the AUC improvement replicates in one transfer direction and is not distinguishable from baseline in the other, while the separability increase itself scales with whether the recipe changes task behavior at all; on uncertainty-weighted fusion – the strongest *unadapted* baseline architecture we find – the same recipe significantly *degrades* performance. We argue this pattern is more consistent with a generic-regularizer account of the mechanism than with literal representation-alignment, since the latter offers no reason for the same objective to reliably help one architecture and reliably hurt another. We argue cross-dataset evaluation should be standard practice for this task, and that auditing easy-to-miss pipeline details – data-loader defaults, split construction, the actual mechanism and architecture-dependence behind an apparent fix – deserves more attention relative to the rate at which new architectures are proposed for these same benchmarks.

I. INTRODUCTION

Pedestrian crossing-intention prediction – estimating, from a short observation window of an ego-vehicle’s onboard sensors, whether a visible pedestrian is about to cross the vehicle’s path – is a safety-critical component of autonomous-driving perception stacks, and has accumulated a substantial architecture literature built around two benchmark datasets, PIE [1] and JAAD [2]. Almost all of this literature, from the original SF-GRU baseline [3] through PCPA [4] to recent transformer- and Mamba-based architectures [5]–[7], reports results trained and tested *within* a single dataset. Gesnouin et al. [8] were, to our knowledge, the first to test the natural follow-up question – does a model trained on one of these datasets generalize to the other? – and found that it largely does not: cross-dataset AUC collapses to near the level of a model with no signal at all. No architecture paper we are aware of published since has re-tested this.

This paper is an extended diagnosis of that collapse, carried out on the SF-GRU architecture, together with one partial, principled fix. Three things motivate treating this as a diagnosis-first paper rather than a methods paper. First, while investigating cross-dataset transfer we discovered that JAAD’s standard data loader silently mislabels the majority of samples under its default configuration (Section III-C): pedestrians lacking a full behavioral annotation are assigned a hard-coded, definitionally uninformative label rather than being excluded, unless the loader’s `sample_type` argument is set explicitly to the literature’s `JAAD_behavior` convention. This is orthogonal to, and independent of, cross-dataset generalization – it affects in-domain JAAD results too – but every cross-dataset result in this paper (and, we suspect, some fraction of prior JAAD-based work using this loader) depends on getting it right, so we treat identifying and correcting it as a contribution in its own right, not merely a footnote to our main experiments.

Second, having corrected the label issue and confirmed the cross-dataset collapse persists, we ran a falsification campaign against the most natural explanations before proposing a fix. PIE and JAAD were filmed with different, largely uncalibrated camera setups; a plausible hypothesis is that cross-dataset failure is substantially a recoverable geometric mismatch. We test this directly with a real metric-space ground-plane reprojection

using PIE’s published camera calibration and researched estimates for JAAD, together with a 20-point sensitivity sweep over JAAD’s unknown camera parameters (Section IV-B), and find no evidence for it: geometric normalization does not recover performance, and no plausible camera-parameter choice for JAAD changes this conclusion. We similarly test, and rule out, video-level train/test leakage as an explanation for a separate, related phenomenon – SF-GRU’s sensitivity to which train/test partition of a dataset is used (Section IV-I). Both are reported as negative results because we think ruling out plausible-sounding explanations is useful independent of whether we found the true cause.

Third, motivated by a modality-zeroing ablation localizing cross-dataset sensitivity to the visual scene-context branch rather than raw trajectory coordinates, we apply domain-adversarial training (a gradient-reversal domain classifier [9]) combined with a contrastive scale-invariance term [10] on the box-trajectory representation. This is, to our knowledge, the first application of domain-adversarial training to cross-dataset pedestrian crossing-intention transfer. It produces a real, seed-confirmed improvement over the uncorrected baseline in both transfer directions (PIE $\rightarrow$ JAAD AUC 0.498 $\rightarrow$ 0.621; JAAD $\rightarrow$ PIE AUC 0.402 $\rightarrow$ 0.613; $n = 6$ seeds each at our best configuration) – but we report this improvement without overstating it: absolute cross-dataset AUC remains far below typical in-domain performance for this model family, and we consider the gap that remains after our best fix to be at least as important a finding as the gap that our fix closes. We therefore probe *why* the fix works, using an independent representation-separability probe distinct from the domain classifier used during training. We find a dissociation, not a correlation, between probe-measured domain separability and task performance (Section IV-E): the configurations that improve cross-dataset AUC also make the fused representation *more*, not less, separable by an independent probe, while a control that reduces separability (same-class mixup) also *reduces* AUC. This suggests domain-adversarial training’s benefit here does not operate through the literal representational invariance it is nominally designed to produce. We further test this recipe on two further, structurally distinct fusion architectures: box-anchored cross-attention (Section III-H) and uncertainty-weighted fusion (Section III-I, chosen because it is the strongest *unadapted* baseline architecture we find). On cross-attention, the AUC improvement holds in one direction and is not distinguishable from baseline in the other; probing both reveals the separability increase itself tracks which direction improves – large where GRL + contrastive helps, negligible where it does not – rather than occurring uniformly regardless of outcome (Section IV-F). On uncertainty-weighted fusion, the recipe does not merely fail to help: it significantly *degrades* the architecture’s strong unadapted baseline in one direction and trends down, non-significantly, in the other (Section IV-G). We treat the combination as evidence for, not against, the generic-regularizer account – a literal representation-alignment mechanism gives no obvious reason for its effect on separability to scale with whether it

helps or hurts the task, nor for the same objective to reliably help one architecture and reliably hurt another – while being explicit that our fix is a reproducible, useful result for SF-GRU specifically and not a general-purpose recipe: of three architectures tested, it helps in both transfer directions on only one.

In summary, this paper’s contributions are:

- 1) Identification and correction of a silent labeling artifact in JAAD’s default data-loading configuration, affecting the large majority of samples in the default (`sample_type='all'`) setting, together with a re-audit of cross-dataset collapse under the corrected labels confirming it is not an artifact of this bug.
- 2) A falsification campaign ruling out camera-geometry mismatch (via real metric-space reprojection and a camera-parameter sensitivity sweep) and video-level train/test leakage as explanations for, respectively, cross-dataset collapse and split-sensitivity.
- 3) A modality-zeroing ablation localizing cross-dataset sensitivity primarily to the visual scene-context modality rather than raw trajectory coordinates.
- 4) A domain-adversarial and contrastive training recipe that partially, reproducibly ($n = 6$ seeds, both directions) mitigates cross-dataset collapse on SF-GRU, together with an honest characterization of how much of the original gap it leaves unclosed.
- 5) A mechanistic probe analysis showing that this recipe’s benefit is dissociated from – not explained by – the representational invariance it is nominally trained to induce, using controls (mixup, a trivial track-length probe, and two further fusion architectures) to support this reading.
- 6) A test of the recipe on two additional, structurally distinct fusion architectures showing it is not a general-purpose fix: it partially replicates on one (helping in one transfer direction) and significantly *harms* the strongest unadapted baseline of the other, evidence we read as further support for the generic-regularizer account over a literal domain-invariance account.

We view this paper’s primary value as diagnostic: a small, well-controlled model family on two well-studied benchmarks, examined closely enough to find a real data bug, rule out two plausible-sounding explanations, and show that a working partial fix does not work for the reason it was designed to. We believe this kind of audit is currently under-supplied relative to the rate at which new architectures are proposed for these same two benchmarks (Section II), none of which re-test cross-dataset transfer.

II. RELATED WORK

A. Architectural Progress on Pedestrian Crossing Intention

PIE [1] and JAAD [2] have been the dominant benchmarks for pedestrian crossing-intention prediction since their introduction, and a substantial body of work has advanced in-dataset accuracy on them through progressively richer

cross-modal fusion: SF-GRU [3] stacks per-modality GRUs; PCPA [4] adds temporal and modality attention over 3D-convolutional, pose, and box streams; more recent work replaces recurrent fusion with transformer- or state-space-based temporal encoders and heavier visual backbones – e.g. Rahman et al. [5] combine a weight-shared Swin Transformer V2 backbone with a Mamba-based motion-encoding module and sparse cross-modal attention (17.23 ms/sample inference; 0.90 AUC on PIE), and TrajFusionNet [6] fuses a sequential branch (trajectory and ego-vehicle speed) with a visual branch (predicted pedestrian bounding boxes overlaid on scene imagery) through a transformer-based architecture. This line of work reports steadily improving in-dataset accuracy and AUC on PIE and JAAD, but – to the best of our knowledge – none of it evaluates cross-dataset generalization; every result we found in this literature is reported under the standard train-test-same-dataset protocol, confirmed by direct inspection of released code where available (e.g. the ADAPT implementation trains and evaluates PIE and JAAD independently, with no cross-dataset transfer).

B. Efficiency-Focused Approaches

A parallel line of work optimizes for inference cost rather than peak accuracy. EfficientPIE [7] argues that a single observed frame, processed by a lightweight convolutional architecture (Fused-MBConv / MBConv blocks with depthwise-separable convolutions), is sufficient for competitive crossing-intention accuracy, reporting 0.21 ms inference latency and state-of-the-art in-dataset accuracy on both PIE (0.92) and JAAD (0.89) – outperforming SF-GRU and every other baseline in their comparison table. This demonstrates that architectural efficiency and competitive in-dataset accuracy are not in tension for this task. However, EfficientPIE’s evaluation, like the architectures in Section II-A, is entirely in-dataset; sole-observation, single-frame architectures in particular raise an open question – not addressed in that work or, to our knowledge, anywhere else – of whether relying on a single frame’s local appearance and context makes a model *more* or *less* sensitive to the dataset-specific visual-context statistics we localize as the dominant driver of cross-dataset sensitivity in Section IV-C.

C. Cross-Dataset and Robustness Evaluation

Gesnoun et al. [8] is, to our knowledge, the only prior work to evaluate pedestrian crossing-intention predictors under a cross-dataset protocol. Training eleven architectures spanning static CNN, 3D-convolutional, pose-based, and RNN-based fusion designs (including an SF-GRU-family model, SFRNN) on one of PIE, JAAD-behavior, or JAAD-all and testing on the others, they find that every architecture generalizes poorly – performance is often statistically indistinguishable from, or worse than, chance – regardless of its in-dataset ranking, and argue that classification metrics alone are insufficient without accompanying uncertainty calibration (Expected and Maximum Calibration Error). They further show that generic ImageNet/Sports1M-pretrained backbones (VGG16,

C3D, I3D) generalize comparably to or better than crossing-intention-specific architectures under domain shift, and that pretraining on data further from the target domain improves calibration under shift. A citation-graph search over all papers citing [8] through 2026 (spanning the architectures in Section II-A onward) found none that independently repeat a PIE↔JAAD cross-dataset evaluation; a 2022 survey of the field cites it as the sole reference for this open problem [11].

Two further papers from the same research group are directly relevant. Achaji et al. [12] show that bounding-box-only features are unexpectedly sufficient for strong in-domain crossing-intention accuracy on PIE, and hypothesize this may reflect a dataset-specific bias rather than a genuinely informative signal; they test this via synthetic-to-real transfer (a CARLA-simulated dataset to PIE) rather than real-to-real cross-dataset transfer, and do not evaluate JAAD. This is, to our knowledge, the closest prior evidence that box-trajectory reliance can reflect a dataset-specific bias rather than a genuinely transferable signal, a hypothesis we test directly with a real cross-dataset, no-box ablation (Section IV-B); our modality-zeroing results (Section IV-C), however, localize cross-dataset sensitivity primarily to the visual scene-context modality rather than to the box trajectory itself, suggesting box-trajectory sufficiency and cross-dataset box-trajectory fragility are related but not identical phenomena for SF-GRU. Gilles et al. [13] study cross-dataset uncertainty for general autonomous-vehicle trajectory forecasting (Argoverse, nuScenes, Interaction, and the Shifts benchmark) rather than pedestrian crossing-intention classification on PIE/JAAD, but establish that cross-dataset evaluation and uncertainty estimation are an active concern in the broader AV-perception community, not one specific to crossing-intention prediction.

Our work directly extends [8]’s finding to the current generation of lightweight fusion architectures: we show the same collapse persists for SF-GRU under corrected JAAD labels (Section IV-A), that it is not resolved by architectural refinements published since 2022 (Section II-A, none of which re-evaluate cross-dataset performance), and that it is not explained by camera-geometry mismatch or video-level split leakage, two plausible confounds neither prior paper tests directly (Sections IV-B and IV-I). Going beyond [8]’s architecture-level comparison and [12]’s in-domain sufficiency finding, we localize the failure to a specific input modality within a single architecture (Section IV-C), apply a domain-adversarial and contrastive mitigation (Section IV-D), and use an independent representation probe to show that mitigation’s benefit is dissociated from the representational invariance it is designed to produce (Section IV-E).

Uncertainty and calibration for pedestrian intention prediction has also been studied outside the cross-dataset setting. Zhang et al. [14] apply evidential deep learning within a transformer to quantify prediction confidence under label noise. Concurrent work on the PSI dataset [15] incorporates a variational bottleneck and a Mahalanobis-distance detector for *within*-dataset out-of-distribution sample detection and selective prediction, improving accuracy by abstaining on the

riskiest individual test samples; this is a complementary but distinct problem from the *between*-dataset domain shift we study, and that work does not evaluate PIE/JAAD or cross-dataset transfer.

D. Positioning

Taken together, the literature on pedestrian crossing-intention prediction has, since [8], continued to report in-dataset accuracy gains through richer fusion (Section II-A) and lower inference cost (Section II-B) without re-examining whether those gains reflect genuine, transferable intention-recognition ability or dataset-specific fit – despite the open problem being visible enough to be cited in a field survey [11]. Our contribution is to re-open that question empirically for the current architecture generation, using a lightweight representative architecture (SF-GRU) for which cross-dataset training and modality-level diagnosis is computationally tractable, and to go beyond diagnosis to a partial, mechanistically-understood fix: a domain-adversarial and contrastive training recipe that reproducibly improves cross-dataset transfer in both directions without achieving the literal representational invariance it is nominally designed around (Section IV-E).

III. METHOD

A. Base Architecture: SF-GRU

We evaluate SF-GRU (Stacked-Fusion GRU) [3], a pedestrian crossing-intention architecture originally proposed by Rasouli et al. SF-GRU processes each input modality with its own single-layer GRU, concatenates that GRU’s output sequence with the raw features of the next modality, and feeds the result into the next GRU in the stack; the final GRU’s last hidden state is passed through a linear layer and sigmoid to produce a crossing probability. We use the following modality order, matching the original paper and its released implementation: local cropped-pedestrian appearance (`local_box`), local surrounding scene context (`local_context`), 2D body pose (`pose`), and raw bounding-box trajectory (`box`). `local_box` and `local_context` are 512-dimensional per-frame features pooled from a frozen, ImageNet-pretrained VGG16 backbone; `pose` is a 36-dimensional per-frame keypoint vector (18 keypoints $\times$ 2D coordinates) from RTMPose [16]; `box` is a 4-dimensional bounding-box coordinate delta. We exclude ego-vehicle speed from our main results (Section III-D) and separately test a synthesized speed signal for JAAD (Section III-K). All modalities are observed over a 15-frame window sampled 60 frames before the annotated crossing/non-crossing event, matching the original protocol’s observation and time-to-event parameters.

We use a PyTorch reimplement of the original Keras/TensorFlow SF-GRU that preserves the model architecture, data-preparation logic (frame sampling, class-balancing via horizontal-flip augmentation of the minority class on the training split only, bounding-box normalization), and training protocol (AdamW, focal binary cross-entropy loss, an F_1 -maximizing decision threshold selected on a held-out valida-

tion split), differing from the original only in framework and in using RTMPose in place of OpenPose for keypoint extraction.

B. Datasets

We use the two datasets SF-GRU was originally evaluated on: PIE [1] (6 hours of continuous ego-vehicle footage, Toronto, Canada; standard split: train = set01+02+04, val = set05+06, test = set03) and JAAD [2] (346 short clips, 2 vehicles, 3 dashcam models, 5 countries; default split with an official train/val/test partition). Both datasets provide per-frame bounding boxes and a per-pedestrian crossing-intention label; we follow each dataset’s standard `seq_type='crossing'` sequence-generation protocol.

C. A Labeling Artifact in JAAD’s Default Configuration

JAAD’s pedestrian identifiers fall into three categories, distinguishable by a substring in the pedestrian ID: bystanders (identifiers containing ‘p’), pedestrians with full behavioral annotation (identifiers containing ‘b’), and a remaining category with neither marker. The dataset loader’s `_get_crossing()` function excludes bystanders explicitly, but for pedestrians in the third, unmarked category it hard-codes both the crossing label and the observation endpoint (`acts=[[0]]`, `event_frame=-1`) regardless of that pedestrian’s actual behavior, because no behavioral annotation exists to determine it. The dataset loader’s public interface exposes a `sample_type` parameter (default ‘all’) that controls whether these unmarked-category pedestrians are included; the standard, published convention – matching the *JAAD_behavior* subset used to report the 686-instance pedestrian count cited in prior work [4] – is `sample_type='beh'`, restricting to only the fully behaviorally-annotated (‘b’) category.

Because `sample_type` defaults to ‘all’ and no downstream training script in the codebase we built on overrides it, every JAAD-side experiment we initially ran silently trained and evaluated against a label distribution in which 72–80% of samples per split (depending on split) carry a label that is not a behavioral observation but a fallback constant. This is not merely severe class imbalance: the label for the majority of samples is definitionally uninformative, independent of any true crossing behavior. We discovered this by noticing an anomalously low, near-constant positive rate (9.2–9.9% across splits) that did not match the literature’s reported *JAAD_behavior* base rates, then confirmed it by direct inspection of the dataset loader’s source and cross-checked with an independent literature audit. Correcting to `sample_type='beh'` raises the positive rate to 62–83% across splits (train 82.6%, val 73.9%, test 62.6%) and reduces the usable JAAD training set from several hundred samples to 195 – a real, unavoidable cost of the fix, since the excluded pedestrians genuinely lack behavioral ground truth and cannot be relabeled. PIE has no equivalent unmarked-pedestrian category and is unaffected. *All results in this paper use the corrected `sample_type='beh'` configuration.* We flag this as a methodological finding independent of our other

results: because this default is silent and the resulting label distribution is not obviously degenerate (a near-10% positive rate is plausible on its face for a crossing-intention task), any prior work built on this dataset loader without explicitly setting `sample_type` is at risk of the same corruption, and we recommend the community audit downstream JAAD-based pipelines for this default.

D. Cross-Dataset Evaluation Protocol

The original SF-GRU evaluation, like nearly all pedestrian crossing-intention work we are aware of prior to Gesnouin et al. [8] and including all 2024–2026 work surveyed in Section II, trains and tests within a single dataset. To assess generalization under dataset shift, we additionally adopt the cross-dataset protocol of [8]: a model is trained (and its decision threshold tuned) on one dataset’s train/val split, then evaluated, unchanged, on the *other* dataset’s held-out test split. We report both directions, $\text{PIE} \rightarrow \text{JAAD}$ and $\text{JAAD} \rightarrow \text{PIE}$.

Because PIE’s `speed` channel is a continuous ego-vehicle OBD sensor reading with no JAAD equivalent – the original implementation substitutes a discrete `vehicle_act` encoding in JAAD’s `speed` slot – we exclude `speed` entirely for our main cross-dataset experiments, following the same exclusion in [8], to avoid conflating a feature-semantics mismatch with genuine domain shift. Our main results therefore use models trained on $\{\text{local_box}, \text{local_context}, \text{pose}, \text{box}\}$.

E. Ruling Out Camera Geometry

A natural hypothesis is that PIE and JAAD’s differing camera installations (PIE: single vehicle, fixed, published intrinsic and extrinsic calibration, including fisheye distortion coefficients; JAAD: multiple vehicles, no published calibration) explain cross-dataset failure through a recoverable geometric mismatch rather than a learned one. We test this directly by projecting each pedestrian’s foot point to ground-plane metric coordinates using PIE’s real published calibration (camera height 1.270 m, pitch -10° , full intrinsic matrix and fisheye distortion model) and researched estimates for JAAD (height 1.40 m, pitch -5° , intrinsics rescaled from PIE’s to JAAD’s frame width), then retraining SF-GRU on the resulting metric-space trajectory instead of raw pixel deltas. We validate the projection independently by checking that the implied real-world pedestrian speed distribution is physically plausible (mean 0.32 m/s, 100% within $[0, 3]$ m/s) before using it for training. We additionally sweep JAAD’s unknown camera height (1.2–1.6 m) and pitch (-3° to -10°) over 20 grid points, reusing a single trained checkpoint’s cached features and varying only the ground-plane projection at evaluation time, to test whether *any* plausible camera-parameter choice for JAAD changes the outcome (Section IV-B).

F. Domain-Adversarial Training

Motivated by ruling out geometry (Section IV-B) and a per-modality zeroing ablation (Section IV-C) that localizes cross-dataset sensitivity to the visual context branch rather than the

raw box trajectory, we train SF-GRU under an unsupervised domain-adaptation (UDA) protocol: a Domain-Adversarial Neural Network (DANN) [9]. A domain classifier, reached from the model’s final fused hidden state through a Gradient Reversal Layer (GRL), is trained to predict which dataset (source or target) a sample came from; the main network is trained, via the reversed gradient, to make that prediction as difficult as possible. The source dataset is fully supervised (crossing labels and domain label); the target dataset’s train-split *inputs only* are used for the domain adversary, its labels and held-out val/test splits never touched during training – the same train-on-source, evaluate-unchanged-on-target protocol as our other cross-dataset experiments, with the addition of unlabeled target inputs available at training time for the domain classifier.

We additionally combine the GRL with a contrastive scale-invariance term: for each source-domain training batch, we synthesize a second “view” of each sample by randomly rescaling its box trajectory around a fixed center (simulating the same physical motion filmed at a different distance or focal length), extract the box modality’s own GRU hidden state for both views, and apply an NT-Xent contrastive loss pulling the two views’ embeddings together while pushing apart other samples in the batch. This trains the box embedding to encode motion pattern rather than absolute pixel scale, independently of the GRL’s broader domain-adversarial pressure on the fused representation. The total training loss is $\mathcal{L} = \mathcal{L}_{\text{crossing}} + \lambda_{\text{dom}}\mathcal{L}_{\text{domain}} + \lambda_{\text{con}}\mathcal{L}_{\text{contrastive}}$, with $\lambda_{\text{dom}} = 1.0$ and the DANN gradient-reversal coefficient following the standard sigmoid ramp schedule of [9]. We report both the a priori default contrastive hyperparameters ($\lambda_{\text{con}} = 0.5$, NT-Xent temperature 0.2) and a small follow-up grid search over $\lambda_{\text{con}} \in \{0.25, 0.5, 1.0\}$, temperature $\in \{0.1, 0.2, 0.4\}$ (Section IV-D), confirming any selected configuration at $n = 6$ seeds before reporting it as a headline number.

G. Representational Invariance Probe

To test whether the GRL’s domain-adversarial training achieves the representational invariance it is designed to produce, we take a trained checkpoint’s final fused hidden state on held-out PIE and JAAD test data and fit a *separate*, freshly-initialized classifier – never touching the original model’s weights – to predict which dataset each representation came from. We report both a linear probe (logistic regression) and a small non-linear probe (2-layer MLP, 32 hidden units, early stopping) to rule out a linear probe being too weak to detect invariant structure a stronger probe would find, averaged over 5 independent random train/test splits of the probe’s own data. This is deliberately distinct from the GRL’s own in-training domain-accuracy diagnostic, which is trained adversarially against a simultaneously changing feature extractor and is not a reliable measure of the final, frozen representation’s actual separability (Section IV-E). We additionally report a trivial control – a probe given only each sample’s track length, a single scalar carrying no pedestrian-behavior information – to

establish how separable PIE and JAAD are from a dataset-fingerprint statistic alone.

H. Second-Architecture Replication

All results above use SF-GRU’s stacked, sequential fusion (each modality’s GRU output concatenated into the next modality’s input, Section III-A). To test whether the GRL + contrastive recipe’s effect and the probe dissociation are specific to this fusion mechanism, we implement the same GRL + contrastive training recipe (Section III-F) on a structurally distinct fusion design: box-anchored cross-attention. Each modality is encoded independently by its own GRU (returning the full hidden-state sequence, not just the final state); the box modality’s encoded sequence serves as the attention query, the concatenation of every other modality’s encoded sequence serves as the key/value context, and the attended output is passed through one further GRU before the classification head – mirroring the box-anchored design used in prior cross-modal-attention variants of SF-GRU, but, to our knowledge, never previously evaluated under the cross-dataset protocol. The GRL and NT-Xent contrastive taps are placed at the analogous points to the stacked-GRU model: the GRL reads the final post-attention GRU’s hidden state (the same representation the classification head reads), and the contrastive term reads the box modality’s own encoder output. We train this architecture at the a priori default contrastive hyperparameters ($\lambda_{\text{con}} = 0.5$, temperature 0.2, Section III-F) rather than re-sweeping for this architecture specifically, since the goal is to test whether the recipe’s effect generalizes, not to find its separately-optimal hyperparameters here; $n = 6$ seeds per direction, matching our other headline confirmations.

I. Third-Architecture Replication

We additionally test a third fusion mechanism: uncertainty-weighted fusion, in which each modality is encoded independently by its own GRU into a single final hidden state (no attention or sequential concatenation), a per-modality log-variance is predicted from that hidden state by a small linear head, and the modalities are combined by a softmax over negative log-variance (an inverse-variance, confidence-weighted average) before the classification head. We choose this architecture specifically, rather than an arbitrary third choice, because it is the strongest *unadapted* baseline in an 8-way cross-dataset architecture sweep we ran across every fusion mechanism implemented in this codebase (PIE $\rightarrow$ JAAD AUC 0.585 ± 0.009 , $n = 3$, versus SF-GRU’s 0.498 ± 0.009) – making it the best-motivated test of whether domain-adversarial training adds value on top of an architecture that already transfers comparatively well without any adaptation. The GRL and contrastive taps read, respectively, the fused (post-uncertainty-weighting) representation and the box modality’s own independent encoder output (available directly here, since each modality’s encoder is independent rather than embedded mid-sequence in a shared stack). Same protocol as Section III-H: default contrastive hyperparameters, $n = 6$ seeds per direction.

J. Pretrained Trajectory Representation

Motivated by Gesnouin et al.’s [8] finding that generic pretrained visual backbones generalize better under PIE/JAAD domain shift than task-specific ones, we test whether the same holds for the box-trajectory modality specifically. We pretrain a small Transformer encoder (2 layers, 4 attention heads, 64-dimensional hidden size) on a masked-frame reconstruction objective: for each box-delta sequence in a pooled, unlabeled corpus (PIE’s and JAAD’s train and validation splits combined, crossing labels never touched), a random subset of frames is replaced with a learned mask token and the encoder must reconstruct each masked frame’s true 4-dimensional delta from the unmasked context, trained with mean-squared-error loss on masked positions only. After pretraining, we freeze the encoder and splice its per-timestep output into SF-GRU’s box GRU slot in place of the raw box delta, training the box GRU (now consuming a 64-dimensional input) and everything downstream of it under the same GRL + contrastive recipe and default hyperparameters as Section III-F, with the pretrained encoder’s weights held fixed throughout. We note explicitly that this is a weaker test of the motivating hypothesis than an ideal version would be, in two independent respects: our pretraining corpus (pooled PIE and JAAD trajectories) is far smaller and less diverse than the external, large-scale motion corpora (e.g. Waymo Open Motion, Argoverse 2) a genuine test would use, *and* it is drawn from the same two datasets used for downstream cross-dataset evaluation, unlike Gesnouin et al.’s source/target-independent visual backbones. We did not have local access to an external motion corpus, and report this as a locally-feasible approximation of the intended experiment, not an equivalent of it (Section IV-H).

K. Synthesizing a Speed Modality for JAAD

PIE’s speed channel is real, continuous ego-vehicle OBD sensor data; JAAD has no equivalent signal, so every experiment above excludes speed from both datasets to keep the modality set comparable, meaning PIE’s model never uses a real signal it natively has. We test whether recovering this modality for JAAD helps, by estimating a coarse per-frame ego-motion proxy from JAAD’s raw video via dense optical flow (Farneback), computed over the region *excluding* the pedestrian’s own bounding box (background motion approximates ego-vehicle motion) and reduced to the per-frame median flow magnitude. We z-score each dataset’s speed channel independently using training-split statistics before use, since PIE’s real km/h-scale readings and JAAD’s unitless optical-flow magnitude are not on a comparable absolute scale. This gives both datasets the full five-modality set (`local_box`, `local_context`, `pose`, `box`, `speed`) rather than the four-modality set used elsewhere in this paper (Section IV-D).

IV. RESULTS

A. Cross-Dataset Collapse Under Corrected Labels

Table I reports in-domain and cross-dataset AUC for an unmodified SF-GRU trained under the corrected JAAD labeling (Section III-C). In-domain, SF-GRU is a competent classifier

TABLE I
IN-DOMAIN AND CROSS-DATASET AUC, UNMODIFIED SF-GRU,
CORRECTED JAAD LABELS. IN-DOMAIN NUMBERS ARE SINGLE-DATASET
TRAIN/TEST (NO DOMAIN SHIFT); CROSS-DATASET NUMBERS USE THE
PROTOCOL OF SECTION III-D. $n = 3$ SEEDS PER CELL.

	In-domain AUC	Cross-dataset AUC
PIE (train/test)	0.78 ± 0.02	–
JAAD (train/test)	0.71 ± 0.03	–
PIE $\rightarrow$ JAAD	–	0.498 ± 0.009
JAAD $\rightarrow$ PIE	–	0.402 ± 0.138

on both datasets. Cross-dataset, performance falls to near-chance in both directions: PIE $\rightarrow$ JAAD AUC 0.498 ± 0.009 and JAAD $\rightarrow$ PIE AUC 0.402 ± 0.138 ($n = 3$ seeds each). The collapse is not an artifact of the labeling bug we identify: it persists, essentially unchanged, after the fix, confirming the finding first reported (under the uncorrected labels) by Gesnoui et al. [8] is real and not an artifact of that prior confound. We note JAAD $\rightarrow$ PIE’s unusually high variance (std 0.138 on $n = 3$) even at baseline, a pattern that recurs across nearly every configuration we test in this direction (Table II) and that we return to in Section IV-K.

B. Ruling Out Geometry and Simple Normalization

Table II (top block) reports three geometry- and scale-motivated fixes, all applied to the raw box/trajectory modality before training: per-dataset box-scale normalization (*scale-norm*), per-dataset pedestrian-height normalization (*height-norm*), and metric-space ground-plane reprojection using each dataset’s camera calibration (*homography*, Section III-E). None recovers baseline performance; several fall *below* the unmodified baseline (e.g. height-norm PIE $\rightarrow$ JAAD AUC 0.390 ± 0.017 , vs. baseline 0.498 ± 0.009). A 20-point grid sweep over JAAD’s unknown camera height (1.2–1.6 m) and pitch (-3° to -10°), re-evaluating a single fixed homography-trained checkpoint at each grid point, found no parameter setting that materially changes cross-dataset AUC (range across the full grid: 0.39–0.43 AUC, PIE $\rightarrow$ JAAD), indicating the null result is not an artifact of a poorly-guessed JAAD camera parameter. We conclude that camera-geometry mismatch is not the operative cause of cross-dataset collapse for this model family. We also test removing the `box` modality entirely (*no-box-speed*, keeping only the appearance/pose/context modalities) as a check on whether raw trajectory coordinates are themselves the leak; this also fails to recover performance and in fact performs worse than the full modality set on PIE $\rightarrow$ JAAD (0.385 vs. 0.498), consistent with the modality-ablation finding in Section IV-C that visual context, not raw box coordinates, drives cross-dataset sensitivity.

C. Modality-Zeroing Ablation

To localize which modality drives cross-dataset sensitivity, we zero one modality’s input at a time (replacing it with its training-set mean) in an otherwise-unmodified trained SF-GRU and re-evaluate cross-dataset. Zeroing `local_context` (the visual scene-context branch)

TABLE II
CROSS-DATASET AUC FOR GEOMETRY/NORMALIZATION FIXES (TOP) AND
DOMAIN-ADAPTATION METHODS (BOTTOM), CORRECTED LABELS, $n = 3$
SEEDS UNLESS NOTED. BEST CONFIGURATION IN BOLD.

Method	PIE $\rightarrow$ JAAD	JAAD $\rightarrow$ PIE
Baseline (Table I)	0.498 ± 0.009	0.402 ± 0.138
Scale-norm	0.437 ± 0.025	0.536 ± 0.063
Height-norm	0.390 ± 0.017	0.554 ± 0.104
Homography (geometry)	0.406 ± 0.011	0.544 ± 0.053
No-box (drop trajectory)	0.385 ± 0.009	0.432 ± 0.085
Contrastive only	0.558 ± 0.028	0.450 ± 0.107
GRL only ($n=6$)	0.608 ± 0.030	0.587 ± 0.069
GRL + contrastive (default)	0.618 ± 0.019	0.583 ± 0.073
GRL + contrastive (tuned, $n=6$)	0.621 ± 0.017	0.613 ± 0.071
+ mixup	0.580 ± 0.077	0.537 ± 0.079
+ synthetic speed	0.602 ± 0.001	0.430 ± 0.072

produces the largest single-modality drop in PIE $\rightarrow$ JAAD AUC (baseline $0.412 \pm 0.007 \rightarrow 0.372 \pm 0.021$) and is the only modality whose removal changes JAAD $\rightarrow$ PIE AUC by more than its own run-to-run noise ($0.545 \pm 0.052 \rightarrow 0.489 \pm 0.085$). Zeroing `pose` changes JAAD $\rightarrow$ PIE AUC by less than one hundredth of a point (a numerical coincidence – the box ablation row below it is the one with real signal – consistent with pose contributing little to this direction, matching our earlier in-domain finding that pose is largely inert here). Zeroing `box` leaves AUC comparatively stable in both directions but collapses PIE $\rightarrow$ JAAD accuracy to 0.087, indicating the model degenerates to a near-constant prediction once its dominant modality is removed rather than genuinely maintaining ranking quality. Taken together with Section IV-B’s null geometry result, this ablation motivated targeting the fused representation broadly (rather than re-encoding the box modality alone) with domain-adversarial training, described next.

D. Domain-Adversarial and Contrastive Training

Table II (bottom block) reports our domain-adaptation results (Section III-F). Contrastive scale-invariance training alone gives a modest, direction-asymmetric improvement (PIE $\rightarrow$ JAAD $0.498 \rightarrow 0.558$; JAAD $\rightarrow$ PIE $0.402 \rightarrow 0.450$, within that direction’s noise). Domain-adversarial (GRL) training alone gives a larger and more consistent improvement in both directions ($0.498 \rightarrow 0.608$; $0.402 \rightarrow 0.587$, $n = 6$). Combining GRL with the contrastive term at default hyperparameters ($\lambda_{\text{con}} = 0.5$, temperature 0.2) slightly exceeds either component alone in both directions. A follow-up grid search over $\lambda_{\text{con}} \in \{0.25, 0.5, 1.0\}$ and temperature $\in \{0.1, 0.2, 0.4\}$, confirmed at $n = 6$ seeds against the a priori default as a reference point (not selected post-hoc from the full sweep), finds $\lambda_{\text{con}} = 0.25$, temperature 0.1 as a small further improvement in both directions: PIE $\rightarrow$ JAAD 0.621 ± 0.017 and JAAD $\rightarrow$ PIE 0.613 ± 0.071 . This is our headline configuration for the remainder of the paper. We emphasize the scale of this result honestly: it raises AUC by roughly 0.12–0.21 over the uncorrected baseline, but remains far below in-domain

performance (Table I) and well short of AUC 0.75–0.85, commonly reported for in-domain SF-GRU-family models. Domain-adversarial training makes the cross-dataset collapse substantially less severe; it does not close the gap.

Two additions to this headline configuration, motivated by hypotheses from elsewhere in the project, both fail to improve it further. Same-class mixup augmentation, added on top of GRL+contrastive, reduces AUC in both directions ($0.621 \rightarrow 0.580$; $0.613 \rightarrow 0.537$) despite – as we show next – *reducing* measured domain separability, the opposite of what a naive account of domain-adversarial training would predict. Synthesized JAAD speed (Section III-K), giving both datasets the full five-modality set, does not improve mean AUC in either direction ($0.621 \rightarrow 0.602$; $0.613 \rightarrow 0.430$) and substantially *worsens* JAAD $\rightarrow$ PIE. We note, without strong interpretation, that the PIE $\rightarrow$ JAAD run with synthetic speed has unusually low variance across seeds (std 0.001 vs. 0.017 for the same configuration without speed); we did not find an explanation for this and flag it as an observation rather than a claim.

E. Representational Separability Is Dissociated From Task Performance

Table III reports linear and non-linear (MLP) probe accuracy at distinguishing PIE from JAAD test-set representations (Section III-G), for the unmodified baseline and for several of the domain-adaptation configurations above. A trivial control – a probe given only each sample’s track length, discarding all learned features – already achieves 84.6% linear accuracy, confirming PIE and JAAD differ in dataset-level statistics detectable without any learned representation at all; all reported probe accuracies should be read relative to this floor, not to chance (50%).

The central finding is a dissociation between separability and task performance. If domain-adversarial training worked by making the fused representation literally domain-invariant, we would expect probe accuracy to *fall* as cross-dataset AUC *rises*. We observe the opposite: every configuration that improves cross-dataset AUC over baseline (full GRL, GRL restricted to the context branch, GRL + contrastive) also *increases* probe accuracy over baseline, both linear (87.7% $\rightarrow$ 95.3–97.1%) and non-linear (82.8% $\rightarrow$ 89.2–94.4%). Conversely, GRL + contrastive + mixup – the one configuration in Table II that *reduces* AUC relative to GRL + contrastive alone – is also the one configuration that *reduces* probe accuracy toward baseline (95.3% $\rightarrow$ 88.3% linear; 89.2% $\rightarrow$ 83.3% MLP). This pattern held across two independent comparisons (the GRL family and the mixup control), suggesting domain-adversarial training’s benefit here is not explained by the literal representational invariance it is nominally trained to produce. We discuss a candidate mechanistic account – that GRL training acts as a generic regularizer on the fused representation, incidentally raising probe-detectable structure while improving task-relevant robustness through a different channel – in Section IV-K.

TABLE III
PIE-VS-JAAD PROBE ACCURACY ON FINAL FUSED HIDDEN STATES (OR, WHERE NOTED, A SINGLE BRANCH’S HIDDEN STATE), AVERAGED OVER 5 RANDOM PROBE TRAIN/TEST SPLITS. HIGHER ACCURACY = MORE SEPARABLE, I.E. LESS DOMAIN-INVARIANT.

Condition	Linear acc.	MLP acc.
Track-length only (trivial)	0.846	0.696 ± 0.272
Baseline (fused)	0.877 ± 0.012	0.828 ± 0.022
GRL only (fused)	0.962 ± 0.017	0.931 ± 0.016
GRL, context branch only	0.948 ± 0.003	0.894 ± 0.038
Baseline, context branch only	0.964 ± 0.009	0.851 ± 0.026
GRL + contrastive (fused)	0.953 ± 0.008	0.892 ± 0.020
+ mixup (fused)	0.883 ± 0.019	0.833 ± 0.013

F. Partial Replication on a Second Architecture

The results above use a single fusion architecture (SF-GRU’s stacked, sequential concat-and-GRU fusion). To test whether the GRL + contrastive recipe’s benefit and the probe dissociation finding are specific to this fusion mechanism or reflect something more general, we build the same GRL + contrastive training recipe (Section III-F) on top of a structurally distinct architecture: box-anchored cross-attention fusion (each modality independently GRU-encoded, the box modality serving as an attention query over the concatenated other modalities, a final GRU over the attended sequence), evaluated cross-dataset for the first time here. At default contrastive hyperparameters (no re-sweep for this architecture; see Section III-F), Table V shows a partial, direction-dependent replication. PIE $\rightarrow$ JAAD improves over the unmodified cross-attention baseline ($0.513 \pm 0.009 \rightarrow 0.559 \pm 0.027$, $n = 6$, a difference that clears the two conditions’ pooled standard deviation), a smaller but real improvement in the same direction as the stacked-GRU result. JAAD $\rightarrow$ PIE does *not* replicate that improvement: the point estimate falls ($0.512 \pm 0.188 \rightarrow 0.446 \pm 0.063$), the opposite sign from the stacked-GRU result on the same direction ($0.402 \rightarrow 0.613$), but the baseline’s own variance in this direction is large enough (std = 0.188, matching the wide JAAD $\rightarrow$ PIE variance noted throughout this paper) that the drop does not clear pooled standard deviation – we can say GRL + contrastive fails to reproduce its PIE $\rightarrow$ JAAD-sized improvement here, but not confidently that it makes this direction significantly worse. We report the point estimate and its uncertainty together rather than rounding either away: the fix’s benefit is not confirmed to be architecture-independent, and a practitioner should not assume it transfers to a new fusion design without re-testing both directions at adequate n .

We probe both directions’ cross-attention checkpoints (Table IV), which reveals a more specific pattern than “GRL + contrastive always increases separability.” For PIE $\rightarrow$ JAAD, the direction that improves, separability rises substantially over the unmodified baseline, both linearly ($0.844 \pm 0.006 \rightarrow 0.967 \pm 0.009$) and non-linearly ($0.846 \pm 0.000 \rightarrow 0.952 \pm 0.020$) – matching every stacked-GRU condition in Table III. For JAAD $\rightarrow$ PIE, the direction where GRL + contrastive fails to reproduce its improvement (its point estimate falls, though

TABLE IV

CROSS-ATTENTION PROBE ACCURACY, BOTH TRANSFER DIRECTIONS. BASELINE USES THE CHECKPOINT TRAINED ON THE ROW’S SOURCE DATASET (PIE-TRAINED FOR THE PIE→JAAD ROW, JAAD-TRAINED FOR JAAD→PIE).

Condition	Linear acc.	MLP acc.
Baseline (PIE-trained)	0.844 ± 0.006	0.846 ± 0.000
GRL + contrastive, PIE→JAAD	0.967 ± 0.009	0.952 ± 0.020
Baseline (JAAD-trained)	0.845 ± 0.004	0.828 ± 0.033
GRL + contrastive, JAAD→PIE	0.851 ± 0.003	0.844 ± 0.004

TABLE V

CROSS-ATTENTION ARCHITECTURE (BOX-ANCHORED FUSION), CORRECTED LABELS, $n = 6$ SEEDS. COMPARE AGAINST TABLE II’S STACKED-GRU NUMBERS FOR THE SAME COMPARISON ON SF-GRU.

Method	PIE→JAAD	JAAD→PIE
Baseline (no adaptation)	0.513 ± 0.009	0.512 ± 0.188
GRL + contrastive	0.559 ± 0.027	0.446 ± 0.063

not by more than the baseline’s own noise; Section IV-F), separability barely moves at all ($0.845 \pm 0.004 \rightarrow 0.851 \pm 0.003$ linear; $0.828 \pm 0.033 \rightarrow 0.844 \pm 0.004$ non-linear) – an order of magnitude smaller shift than the PIE → JAAD case. Separability and task performance are therefore not simply decoupled in the sense of moving in opposite directions; rather, the training procedure’s measurable effect on the representation tracks whether it helps the task at all. Where GRL + contrastive meaningfully changes cross-dataset AUC (for better, on stacked-GRU in both directions and cross-attention PIE → JAAD), it also meaningfully changes probe separability, always upward, never downward; where it does not help (cross-attention JAAD → PIE), it leaves the representation close to where the unmodified baseline left it, and the AUC change in that case is presumably driven by something else in training (e.g. optimization dynamics specific to this architecture and direction) rather than by any representational shift this probe can detect. This refines, rather than weakens, our reading in Section IV-E: literal domain-invariance is not what GRL + contrastive achieves in any condition we test, and the magnitude of its (upward) effect on separability appears coupled to whether it changes task behavior at all, not decoupled from it as a simpler “always more separable” story would suggest.

G. A Third Architecture: The Recipe Can Actively Hurt

We test a third, again structurally distinct architecture: uncertainty-weighted fusion, in which each modality is independently GRU-encoded and fused by a softmax over per-modality inverse-variance weights (a learned confidence-weighted average) rather than sequential concatenation (SF-GRU) or query-anchored attention (cross-attention). This architecture is not an arbitrary third choice: it is the strongest *unadapted* baseline we find in an 8-way cross-dataset architecture sweep (PIE → JAAD AUC 0.585 ± 0.009 , $n = 3$, exceeding SF-GRU’s 0.498 ± 0.009 and cross-attention’s 0.513 ± 0.009

TABLE VI

UNCERTAINTY-WEIGHTED FUSION ARCHITECTURE, CORRECTED LABELS. BASELINE IS $n = 3$ (FROM THE 8-WAY ARCHITECTURE SWEEP); GRL + CONTRASTIVE IS $n = 6$. ↓* = DROP CLEARS POOLED STANDARD DEVIATION; ↓ = POINT ESTIMATE FALLS BUT NOT SIGNIFICANTLY AT THIS n .

Method	PIE→JAAD	JAAD→PIE	Verdict
Baseline (no adaptation)	0.585 ± 0.009	0.377 ± 0.046	–
GRL + contrastive	0.499 ± 0.072	0.351 ± 0.060	↓*/↓

by a wide margin), making it the best-motivated candidate for testing whether domain-adversarial training adds value on top of an architecture that already generalizes comparatively well without any adaptation. We apply the same GRL + contrastive recipe, default hyperparameters, box-embedding contrastive tap on this architecture’s own independent box encoder output, $n = 6$ seeds per direction.

Table VI shows the recipe does not help here, and on PIE → JAAD actively *hurts*: AUC falls from 0.585 ± 0.009 to 0.499 ± 0.072 , a drop that clears the two conditions’ pooled standard deviation and is the only case in this paper where GRL + contrastive significantly *underperforms* an architecture’s own unadapted baseline. JAAD → PIE also falls ($0.377 \pm 0.046 \rightarrow 0.351 \pm 0.060$), though this difference does not clear pooled standard deviation. Combined with the cross-attention result (Section IV-F), only one of three architectures we test (SF-GRU) shows the recipe helping in *both* transfer directions; cross-attention shows it helping in one and not distinguishable from baseline in the other; uncertainty-weighted fusion shows it significantly hurting in one and not distinguishable from baseline (trending down) in the other. We revise our framing accordingly: the earlier characterization of this recipe as producing a “partial, principled fix” understates how architecture-contingent it is. A more accurate summary is that GRL + contrastive is a reproducible fix for SF-GRU’s specific stacked-fusion topology, whose mechanism (per Sections IV-E and IV-F) does not appear to be literal representational alignment, and which does not transfer as a recommended recipe to the two alternative fusion mechanisms we test – one of which it actively harms on its strongest baseline.

H. A Pretrained Trajectory Representation Does Not Help

Gesnoui et al. [8] report that generic, ImageNet/Sports1M-pretrained visual backbones generalize better under PIE/JAAD domain shift than crossing-intention-specific visual architectures – motivating us to test the analogous question for the *motion* modality: does a box-trajectory representation pre-trained on a broader distribution than either dataset’s own labeled training pool generalize better cross-dataset than SF-GRU’s from-scratch box GRU? Lacking access to a large external trajectory corpus, we pretrain a small Transformer encoder with a masked-frame reconstruction objective (BERT-style: random frames of a box-delta sequence are masked and predicted from context) on the pooled, *unlabeled* box-delta trajectories from both PIE’s and JAAD’s train and validation

splits combined – a broader motion distribution than either dataset’s own labeled training pool alone, though a materially weaker substitute for a genuine large-scale external corpus (e.g. Waymo Open Motion or Argoverse 2), which we did not have local access to. We then freeze this encoder and splice its per-timestep output into SF-GRU’s box GRU slot in place of the raw 4-dimensional box delta, training only the downstream fusion (the box GRU, now consuming a 64-dimensional embedding, and everything after it) under the same GRL + contrastive recipe and default hyperparameters as Section IV-D.

Table VII shows this does not help, and in fact underperforms both the from-scratch GRL + contrastive result and, in both directions, the plain unadapted baseline (Table I: 0.498 and 0.402). We do not read this as evidence against a pretrained-backbone strategy in general, and we are explicit about a confound in this experiment beyond corpus size alone: our pretraining pool is drawn from PIE’s and JAAD’s *own* train and validation splits, the same two datasets used for the downstream cross-dataset evaluation, whereas the visual-backbone analogy this experiment is motivated by (Gesnouin et al.) concerns a backbone pretrained on data with no relationship to either evaluation domain. A self-supervised objective trained on a mixture of the source and target domains has no pressure to discard whatever dataset-identifying structure (frame rate, box-annotation convention, scale) distinguishes the two – indeed, our own probe methodology (Section III-G) shows PIE and JAAD are separable from nearly any representation of their inputs, including a single scalar track length. It is therefore not clear whether this experiment tested “does motion pretraining help cross-dataset transfer” or something closer to “does an encoder pretrained to reconstruct a mixture of both domains’ own trajectories help,” which is a different, and less informative, question. Corpus size (a few hundred trajectories, vs. ImageNet’s millions of images) compounds this: unlike a frozen VGG16 backbone pretrained on a large, source/target-independent corpus, a box encoder pretrained on a small pool drawn from the very datasets under study does not obviously carry information beyond what the from-scratch, task-supervised box GRU already has access to, and the added representation bottleneck (a frozen, non-task-tuned embedding standing between the raw trajectory and the classification objective) may cost more than the pretraining gives back at this scale. Isolating which of these two factors – domain composition or corpus size – drives the negative result (for instance, by pretraining on PIE alone and evaluating JAAD → PIE, or vice versa, so the pretraining corpus excludes the source domain entirely) is a direction for future work we did not pursue here.

I. Split Sensitivity Is Not Explained by Video-Level Leakage

SF-GRU-family cross-dataset results are also sensitive to which train/val/test partition is used within a dataset. Under JAAD’s official default split, our GRL + contrastive model reaches PIE → JAAD AUC 0.621 ± 0.017 ; under a random per-pedestrian partition (`data_split_type='random'`,

TABLE VII
FROZEN PRETRAINED TRAJECTORY ENCODER SPLICED INTO THE BOX MODALITY, GRL + CONTRASTIVE RECIPE, DEFAULT HYPERPARAMETERS, CORRECTED LABELS, $n = 6$ SEEDS. COMPARE AGAINST TABLE II’S FROM-SCRATCH-BOX GRL + CONTRASTIVE ROW (DEFAULT HYPERPARAMETERS: $0.618 \pm 0.019 / 0.583 \pm 0.073$).

Method	PIE→JAAD	JAAD→PIE
GRL + contrastive, pretrained traj. box	0.489 ± 0.019	0.440 ± 0.089

ratios $[0.5, 0.4, 0.1]$, freshly redrawn per run), the same method scores 0.511 ± 0.036 – a substantial drop. Because a per-pedestrian random split does not guarantee that all pedestrians from a given video stay in the same split, we hypothesized this drop was explained by train/test video overlap: the default random split places 52% of JAAD pedestrians and (via a pedestrian-count coincidence) effectively 100% of PIE pedestrians in videos that appear on both sides of the train/test boundary. We built a video-grouped variant of the random split (Section III-G’s counterpart for splitting, grouping all pedestrians in a video before partitioning, guaranteeing zero video overlap) and re-ran the same method under it. Contrary to our hypothesis, video-grouped-split AUC is *lower*, not higher, than the leaky pedestrian-level random split: 0.445 ± 0.054 ($n = 3$) vs. 0.511 ± 0.036 . Video-level leakage is therefore not the explanation for the default-split-vs-random-split gap; the true cause remains open. We report this as a negative result rather than discarding it, since it rules out what seemed the most likely explanation and narrows the space of remaining hypotheses (e.g. the default splits may differ systematically in scene composition or pedestrian-behavior distribution in a way a random partition does not control for).

J. Summary of Everything Attempted

Table VIII collects every fix attempt in this paper in one place, since the individual results are otherwise spread across seven subsections written around different hypotheses, with a verdict computed against pooled standard deviation rather than by eye. Of the twelve non-baseline interventions, only the domain-adversarial family (GRL alone, GRL + contrastive) improves PIE → JAAD *and* JAAD → PIE on the primary architecture (SF-GRU) by more than pooled noise in both directions. The geometry- and normalization-based fixes (rows 2–5) all significantly *hurt* PIE → JAAD and are statistically indistinguishable from baseline on JAAD → PIE – not the symmetric two-direction harm a first read of the raw numbers might suggest, since JAAD → PIE’s wide baseline variance (row 1; Section IV-K) absorbs differences that would be significant in the other direction. Synthetic speed and the pretrained trajectory representation (rows 10–11) both significantly hurt *both* directions relative to the tuned headline recipe; mixup (row 9) is statistically indistinguishable from it in both directions despite its lower point estimate. Applied to the two alternative fusion architectures (rows 12–13), the recipe’s record is mixed-to-negative: on cross-attention it significantly helps PIE → JAAD and is statistically indistinguishable from

TABLE VIII

EVERY INTERVENTION ATTEMPTED IN THIS PAPER, BOTH TRANSFER DIRECTIONS, CORRECTED LABELS. SF-GRU (STACKED FUSION) UNLESS NOTED; ROW 4 (HOMOGRAPHY) IS THE METRIC GROUND-PLANE REPROJECTION OF SECTION III-E, ROW 5 DROPS THE BOX MODALITY ENTIRELY. VERDICT IS $\uparrow/\downarrow/\approx$ RELATIVE TO THE STATED BASELINE, WHERE $\approx$ MEANS THE DIFFERENCE IS SMALLER THAN THE TWO CONDITIONS’ POOLED STANDARD DEVIATION (I.E. NOT DISTINGUISHABLE AT THIS n) – ROWS 2–8 COMPARE AGAINST THE UNADAPTED BASELINE (ROW 1); ROWS 9–11 COMPARE AGAINST GRL + CONTRASTIVE TUNED (ROW 8); ROWS 12–13 COMPARE AGAINST THEIR OWN ARCHITECTURE’S UNADAPTED BASELINE (TABLES V AND VI).

Intervention	PIE→JAAD	JAAD→PIE
1. Baseline (no fix)	0.498 $\pm$ 0.009	0.402 $\pm$ 0.138
2. Scale-norm	0.437 $\pm$ 0.025	0.536 $\pm$ 0.063
3. Height-norm	0.390 $\pm$ 0.017	0.554 $\pm$ 0.104
4. Homography	0.406 $\pm$ 0.011	0.544 $\pm$ 0.053
5. No-box	0.385 $\pm$ 0.009	0.432 $\pm$ 0.085
6. Contrastive only	0.558 $\pm$ 0.028	0.450 $\pm$ 0.107
7. GRL only	0.608 $\pm$ 0.030	0.587 $\pm$ 0.069
8. GRL+contrastive (tuned)	0.621 $\pm$ 0.017	0.613 $\pm$ 0.071
9. + mixup	0.580 $\pm$ 0.077	0.537 $\pm$ 0.079
10. + synth. speed	0.602 $\pm$ 0.001	0.430 $\pm$ 0.072
11. + pretrained traj.	0.489 $\pm$ 0.019	0.440 $\pm$ 0.089
12. Cross-attn.	0.559 $\pm$ 0.027	0.446 $\pm$ 0.063
13. Uncertainty fusion	0.499 $\pm$ 0.072	0.351 $\pm$ 0.060

its own baseline (not significantly worse) on JAAD $\rightarrow$ PIE; on uncertainty-weighted fusion – the strongest unadapted baseline of any architecture we test – it significantly *hurts* PIE $\rightarrow$ JAAD and trends down (not significant) on JAAD $\rightarrow$ PIE. Only SF-GRU shows the recipe helping in both directions. We read this table as the paper’s most concise evidence for its central claim: cross-dataset collapse in this model family is not fixed by any single targeted intervention we tried except domain-adversarial training on SF-GRU specifically, and that fix is not just partial but architecture- contingent, actively harmful on at least one alternative architecture’s strongest baseline.

K. Discussion

Four results together shape our interpretation. First, geometry- and normalization-based fixes are uniformly ineffective (Section IV-B), ruling out the most straightforward explanation (differing camera setups) for cross-dataset collapse. Second, the modality-ablation result (Section IV-C) and the probe results (Section IV-E) together point away from a single-modality, single-cause account: visual context matters most to zero out, yet the mechanism that helps most (GRL) does not work by suppressing a detectable dataset signal – it *increases* that signal’s detectability while improving task performance. We read this as evidence that GRL’s practical benefit in this setting is closer to a generic regularization effect on the fused representation (e.g. reducing reliance on dataset-specific spurious correlations the main task loss alone would otherwise fit) than to the literal representation-alignment mechanism it is designed around, echoing similar decoupling results reported for DANN-family methods outside this domain. Third, the cross-attention replication (Section IV-F) sharpens this reading

further: probing both transfer directions shows the separability increase is not a fixed side effect of the training procedure, but tracks whether GRL + contrastive changes task behavior at all – large where it helps (PIE $\rightarrow$ JAAD on both architectures, JAAD $\rightarrow$ PIE on stacked-GRU) and negligible where it does not (JAAD $\rightarrow$ PIE on cross-attention). A representation-alignment account predicts a consistent effect on separability regardless of whether AUC improves; what we observe – the separability shift and the AUC shift rising and falling together in magnitude, even though the AUC shift is not always

positive – is more consistent with GRL + contrastive perturbing the representation only when it meaningfully engages with training at all, and that engagement sometimes helping and sometimes hurting the task depending on architecture-specific training dynamics. The uncertainty-weighted fusion result (Section IV-G) is the sharpest data point for this view: applied to the architecture with the *best* unadapted baseline of any we test, the same recipe significantly *degrades* it. If GRL + contrastive’s benefit came from a mechanism intrinsic to domain-adversarial training working correctly, we would not expect it to reliably help one architecture and reliably hurt another; a generic-regularizer account, where the training procedure’s effect on a given architecture’s specific optimization landscape can go either way, accommodates this more naturally than an account in which the method’s success is attributed to achieving its nominal objective. Fourth, the persistent, large JAAD $\rightarrow$ PIE variance across nearly every method in Tables II, V, and VI deserves an explicit alternative explanation before we attribute it to the benchmark’s scene composition: JAAD’s default-split training set has 103 labeled sequences, against PIE’s 796 – an 8-fold difference – so JAAD-as-source runs are trained on a much smaller, and therefore individually noisier, sample than PIE-as-source runs, quite apart from any property of *which* scenes each dataset contains. A smaller source training set producing a noisier decision boundary, and hence higher-variance transfer to any target, is a mundane statistical explanation that competes directly with a benchmark-composition account, and our experiments do not distinguish between them: every JAAD-as-source condition in this paper confounds “JAAD” with “the smaller-training-set direction.” The counter-hypothesis split-sensitivity result (Section IV-I) is consistent with either account. We flag this rather than resolve it: a clean test would hold training-set size fixed (e.g. subsampling PIE’s source training set to JAAD’s size) and check whether PIE $\rightarrow$ JAAD variance rises to match, which we did not run. Together, this and the split-sensitivity result suggest that some of this task’s difficulty is a property of the benchmark itself – scene composition, training-set size, or both – rather than purely a property of any model or training method evaluated on it. We view robust cross-dataset pedestrian intent prediction as still substantially unsolved, with our contribution best read as (a) a fix that is reproducible and genuinely useful for one specific architecture (SF-GRU’s stacked fusion) but is not a general-purpose recipe – tested on two further, structurally distinct architectures, it helps partially on one and significantly hurts the other’s

best baseline in at least one direction – together with a mechanistic account (Sections IV-E–IV-G) of why its effect is architecture-contingent rather than a property of whether it achieves genuine domain-invariance, and (b) a set of ruled-out hypotheses – geometry, video-level leakage, a missing speed modality, and a locally-pretrained trajectory representation – that we hope narrows the search space for future work more than our single positive result advances it.

V. CONCLUSION

We set out to understand why SF-GRU-family pedestrian crossing-intention models fail to generalize across PIE and JAAD, and found more than one answer was needed. Part of the difficulty was a silent, orthogonal data bug in JAAD’s default configuration, unrelated to cross-dataset transfer itself but affecting any result built on the unaudited default; we identify, fix, and re-confirm our findings hold under the correction. The remaining, genuine cross-dataset collapse is not explained by camera geometry, and the closely related phenomenon of split-sensitivity is not explained by video-level train/test leakage – both plausible, natural hypotheses that we falsify directly rather than merely fail to support. A domain-adversarial and contrastive training recipe, applied to the fused representation and motivated by a modality-zeroing ablation, produces a real, seed-confirmed improvement in both transfer directions on SF-GRU, but leaves cross-dataset performance well below in-domain levels; we consider this residual gap, not the improvement, to be the more important number to report honestly. A representation-separability probe, deliberately independent of the domain classifier used during training, shows this recipe’s benefit is dissociated from the literal domain-invariance it is designed to induce – the representations that help most are, if anything, *more* separable by an external probe, not less – suggesting its practical effect is closer to a generic regularizer on the fused representation than to representation alignment per se. Testing the same recipe on two further, structurally distinct fusion architectures supports this reading further and tempers any claim of a general-purpose fix: on cross-attention fusion the improvement replicates only partially, and on uncertainty-weighted fusion – the strongest unadapted baseline architecture we find – the same recipe significantly *degrades* performance. We take from this that domain-adversarial training’s value here is real but architecture-specific, not evidence that it achieves anything close to its nominal objective of representational invariance regardless of architecture.

We draw two broader conclusions. First, cross-dataset evaluation should be treated as standard practice for pedestrian crossing-intention architectures, not an optional follow-up: every architecture we survey in Section II published since Gesnoui et al.’s original finding [8] reports only in-domain results, and our own in-domain numbers (Table I) look entirely unremarkable next to a near-chance cross-dataset collapse the same checkpoint exhibits. In-domain performance on PIE or JAAD alone is not informative about whether a proposed architecture has learned anything that transfers. Second, we

think auditing benchmark and pipeline details that are easy to get silently wrong – data-loader defaults, split construction, the provenance of a supposedly domain-invariant representation – is undersupplied relative to the rate at which new architectures are proposed for these benchmarks, and that the negative results in this paper (ruled-out geometry, ruled-out leakage, a dissociated invariance mechanism) are worth reporting even though none of them, individually, is the paper’s headline number.

Limitations. Our experiments center on SF-GRU’s stacked fusion; we additionally test our headline recipe on two further, structurally distinct architectures (box-anchored cross-attention, Section IV-F, and uncertainty-weighted fusion, Section IV-G). The probe dissociation – higher separability wherever the recipe changes task behavior at all – replicates on both, but the *fix itself* does not: cross-attention shows a partial, one-direction improvement, and uncertainty-weighted fusion shows the recipe significantly harming its (strongest, of any architecture we test) unadapted baseline. We therefore have reasonable evidence the dissociation mechanism is not an SF-GRU-specific artifact, but active evidence *against* the fix being architecture-general, not merely an absence of evidence for it. We did not test transformer- or Mamba-based architectures at all, so whether our findings extend to substantially larger or differently-inductive-biased fusion mechanisms remains open, and a practitioner adopting this recipe on a new architecture should re-validate both directions rather than assume transfer. We evaluate on two datasets; both are pedestrian-crossing-specific, relatively small by modern standards, and share enough protocol lineage (both from the same research group) that a third, more independently sourced dataset (e.g. TITAN) would strengthen any claim that our findings reflect a general property of this task rather than a property specific to the PIE/JAAD pair. Our camera-geometry falsification relies on a researched, not measured, camera calibration for JAAD, since JAAD does not publish one; we mitigate this with a sensitivity sweep over plausible parameter ranges but cannot rule out that a parameter setting outside our swept range behaves differently. Finally, our synthesized JAAD speed modality is a coarse optical-flow proxy, not a calibrated measurement, and its negative result should be read as evidence about this particular proxy rather than as a general claim that a speed signal cannot help JAAD cross-dataset transfer.

We release our corrected-labeling configuration, falsification experiments, and probe methodology in the hope that they are directly reusable by future work evaluating new architectures on PIE and JAAD, independent of whether that work adopts our specific training recipe.

REFERENCES

- [1] A. Rasouli, I. Kotseruba, T. Kunic, and J. K. Tsotsos, “Pie: A large-scale dataset and models for pedestrian intention estimation and trajectory prediction,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6262–6271.
- [2] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, “Are they going to cross? a benchmark dataset and baseline for pedestrian crosswalk behavior,” in *Proceedings of the IEEE International Conference on Computer Vision Workshops*, 2017, pp. 206–213.

- [3] —, “Pedestrian action anticipation using contextual feature fusion in stacked rnns,” *arXiv preprint arXiv:2005.06582*, 2020, bMVC 2019.
- [4] I. Kotseruba, A. Rasouli, and J. K. Tsotsos, “Benchmark for evaluating pedestrian action prediction,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 1258–1268.
- [5] M. M. Rahman, P. Zhou, A. F. M. A. Noor, M. I. Ahasan, K. O. M. Goh, S. M. H. Mahmud, M. M. Rahman, and K. Gao, “Adaptive cross-modal fusion with sparse attention for pedestrian crossing intention prediction,” *arXiv preprint arXiv:2607.12293*, 2026, under review at PeerJ Computer Science.
- [6] F. G. Landry and M. A. Akhloufi, “Trajfusnet: Pedestrian crossing intention prediction via fusion of sequential and visual trajectory representations,” *arXiv preprint arXiv:2508.19866*, 2026, submitted to IEEE Transactions on Intelligent Vehicles.
- [7] F. Qu, P. Zhou, Y. He, K. Gao, Y. Luo, X. Feng, Y. Liu, and S. Guo, “Efficientpie: Real-time prediction on pedestrian crossing intention with sole observation,” in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI-25)*, 2025, pp. 1793–1801.
- [8] J. Gesnouin, S. Pechberti, B. Stanciulescu, and F. Moutarde, “Assessing cross-dataset generalization of pedestrian crossing predictors,” *arXiv preprint arXiv:2201.12626*, 2022.
- [9] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, “Domain-adversarial training of neural networks,” *Journal of Machine Learning Research*, vol. 17, no. 59, pp. 1–35, 2016, arXiv:1505.07818.
- [10] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020, pp. 1597–1607.
- [11] J. Fang, F. Wang, P. Shen, Z. Zheng, J. Xue, and T.-S. Chua, “Behavioral intention prediction in driving scenes: A survey,” *arXiv preprint arXiv:2211.00385*, 2022, published version: IEEE Transactions on Intelligent Transportation Systems, 2024.
- [12] L. Achaji, J. Moreau, T. Fouqueray, F. Aioun, and F. Charpillet, “Is attention to bounding boxes all you need for pedestrian action prediction?” *arXiv preprint arXiv:2107.08031*, 2022, iIEEE Intelligent Vehicles Symposium (IV) 2022.
- [13] T. Gilles, S. Sabatini, D. Tsishkou, B. Stanciulescu, and F. Moutarde, “Uncertainty estimation for cross-dataset performance in trajectory prediction,” *arXiv preprint arXiv:2205.07310*, 2022, iCRA 2022 Workshop on Fresh Perspectives on the Future of Autonomous Driving.
- [14] Z. Zhang, R. Tian, and Z. Ding, “Trep: Transformer-based evidential prediction for pedestrian intention with uncertainty,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 3, 2023, pp. 3534–3542.
- [15] S. Ashayer, H. H. Nguyen, Y. Liang, and M. Sartipi, “Pedestrian crossing intent prediction via psychological features and transformer fusion,” *arXiv preprint arXiv:2603.19533*, 2026.
- [16] T. Jiang, P. Lu, L. Zhang, N. Ma, R. Han, C. Lyu, Y. Li, and K. Chen, “RtmPose: Real-time multi-person pose estimation based on mmpose,” *arXiv preprint arXiv:2303.07399*, 2023.